

CHAPTER 2

Human Reproduction

Male Reproductive System

- Select the correct sequence for transport of sperm cells in male reproductive system. (2019)
 - Testis → Epididymis → Vasa efferentia → Rete testis → Inguinal canal → Urethra
 - Seminiferous tubules → Rete testis → Vasa efferentia → Epididymis → Vas deferens → Ejaculatory duct → Urethra → Urethral meatus
 - Seminiferous tubules → Vasa efferentia → Epididymis → Inguinal canal → Urethra
 - Testis → Epididymis → Vasa efferentia → Vas deferens → Ejaculatory duct → Inguinal canal → Urethra → Urethral meatus
- Which of the following depicts the correct pathway of transport of sperms? (2016 - II)
 - Rete testis → Vas deferens → Efferent ductules → Epididymis
 - Efferent ductules → Rete testis → Vas deferens → Epididymis
 - Rete testis → Efferent ductules → Epididymis → Vas deferens
 - Rete testis → Epididymis → Efferent ductules → Vas deferens
- The shared terminal duct of the reproductive and urinary system in the human male is: (2015, 14)
 - Urethra
 - Ureter
 - Vas deferens
 - Vasa efferentia

Female Reproductive System

- Arrange the following parts in human Mammary gland, traversing the route of milk ejection. (2024 Re)
 - Mammary duct
 - Lactiferous duct
 - Mammary alveolus
 - Ampulla
 - Mammary tubule
 Choose the **correct** answer from the options given below:
 - D → C → E → A → B
 - C → E → B → A → D
 - C → E → A → D → B
 - A → C → E → D → B
- Match **List-I** with **List-II** relating to human female external genitalia: (2024 Re)

List-I (Structures)		List-II (Features)	
(A)	Mons pubis	(I)	A fleshy fold of tissue surrounding the vaginal opening
(B)	Clitoris	(II)	Fatty cushion of cells covered by skin and hair

(C)	Hymen	(III)	Tiny finger-like structure above labia minora
(D)	Labia majora	(IV)	A thin membrane-like structure covering vaginal opening

Choose the **correct** answer from the options given below:

- A-II, B-III, C-IV, D-I
 - A-IV, B-III, C-II, D-I
 - A-I, B-IV, C-III, D-II
 - A-II, B-III, C-I, D-IV
- Given below are two statements: (2024)

Statement I: The presence or absence of hymen is not a reliable indicator of virginity.

Statement II: The hymen is torn during the first coitus only.

 In the light of the above statements, choose the correct answer from the options given below :
 - Statement I is true but Statement II is false
 - Statement I is false but Statement II is true
 - Both Statement I and Statement II are true
 - Both Statement I and Statement II are false
 - Which of the following is not a component of Fallopian tube? (2024)
 - Infundibulum
 - Ampulla
 - Uterine fundus
 - Isthmus
 - Given below are two statements: (2023)

Statement I: Vas deferens receives a duct from seminal vesicle and opens into urethra as the ejaculatory duct.

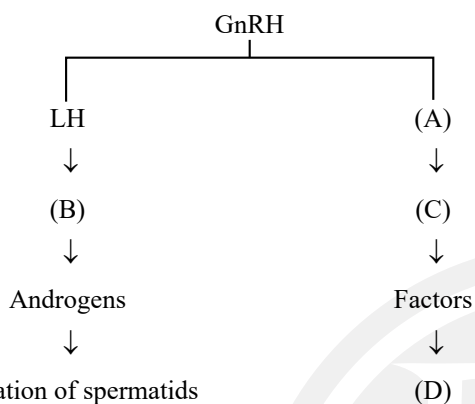
Statement II: The cavity of the cervix is called cervical canal which along with vagina forms birth canal.

 In the light of the above statements, choose the correct answer from the options given below:
 - Statement I incorrect but Statement II is true.
 - Both Statement I and Statement II are true.
 - Both Statement I and Statement II are false.
 - Statement I is correct but Statement II is false.
 - Arrange the components of mammary gland (from proximal to distal). (2022 Re)
 - Mammary duct
 - Lactiferous duct
 - Alveoli
 - Mammary ampulla
 - Mammary tubules
 Choose the most appropriate answer from the options given below.
 - E → C → D → B → A
 - C → A → D → E → B
 - B → C → E → D → A
 - C → E → A → D → B

10. Capacitation occurs in: (2017-Delhi)
 a. Rete testis b. Epididymis
 c. Vas deferens d. Female Reproductive tract
11. Capacitation refers to changes in the: (2015)
 a. Ovum after fertilisation b. Sperm after fertilisation
 c. Sperm before fertilisation d. Ovum before fertilisation

Gametogenesis

12. Identify the correct option (A), (B), (C), (D) with respect to spermatogenesis. (2024)



- a. FSH, Sertoli cells, Leydig cells, spermatogenesis.
 b. ICSH, Leydig cells, Sertoli cells, spermatogenesis.
 c. FSH, Leydig cells, Sertoli cells, spermiogenesis
 d. ICSH, Interstitial cells, Leydig cells, spermiogenesis.
13. Arrange the sequence of different hormones for their role during gametogenesis. (2023-Manipur)
 A. Gonadotropin LH stimulates synthesis and secretion of Androgen
 B. Gonadotropin releasing hormone from hypothalamus
 C. Androgen stimulates spermatogenesis
 D. Gonadotropin FSH helps in the process of spermiogenesis
 E. Gonadotropins from anterior pituitary gland.
 Choose the **correct** answer from the options given below:
 a. (E), (A), (D), (B), (C) b. (C), (A), (D), (E), (B)
 c. (B), (E), (A), (C), (D) d. (D), (B), (A), (C), (E)

14. Given below are two statements regarding oogenesis: (2023-Manipur)

Statement I: The primary follicles get surrounded by more layers of granulosa cells, a theca and shows fluid filled cavity antrum. Now it is called secondary follicle.

Statement II: Graffian follicle ruptures to release the secondary oocyte from the ovary by the process called ovulation.

In the light of the above statements, choose the **correct** answer from the options given below:

- a. **Statement I** is correct but **Statement II** is incorrect.
 b. **Statement I** is incorrect but **Statement II** is correct.
 c. Both **Statement I** and **Statement II** are correct.
 d. Both **Statement I** and **Statement II** are incorrect.
15. How many secondary spermatocytes are required to form 400 million spermatozoa? (2022 Re)
 a. 400 million b. 50 million
 c. 100 million d. 200 million

16. Which of the following statements are true for spermatogenesis but do not hold true for Oogenesis? (2022)

- A. It results in the formation of haploid gametes
 B. Differentiation of gamete occurs after the completion of meiosis
 C. Meiosis occurs continuously in a mitotically dividing stem cell population
 D. It is controlled by the Luteinising hormone (LH) and Follicle Stimulating Hormone (FSH) secreted by the anterior pituitary
 E. It is initiated at puberty

Choose the most appropriate answer from the options given below.

- a. B, C and E only b. C and E only
 c. B and C only d. B, D and E only

17. At which stage of life the oogenesis process is initiated? (2022)

- a. Adult
 b. Puberty
 c. Embryonic development stage
 d. Birth

18. Given below are two statements: (2022)

Statement I: The release of sperms into the seminiferous tubules is called spermiation

Statement II: Spermiogenesis is the process of formation of sperms from spermatogonia

In the light of the above statements, choose the most appropriate answer from the options given below.

- a. Statement I is incorrect but Statement II is correct
 b. Both Statement I and Statement II are correct
 c. Both statement I and statement II are incorrect
 d. Statement I is correct but Statement II is incorrect

19. Meiotic division of the secondary oocyte is completed: (2020)

- a. At the time of copulation
 b. After zygote formation
 c. At the time of fusion of a sperm with an ovum
 d. Prior to ovulation

20. Select the correct option of haploid cells from the following groups: (2020-Covid)

- a. Secondary spermatocyte, First polar body, Ovum
 b. Spermatogonia, Primary spermatocyte, Spermatid
 c. Primary spermatocyte, Secondary spermatocyte, Second polar body
 d. Primary oocyte, Secondary oocyte, Spermatid

21. Extrusion of second polar body from egg nucleus occurs: (2019)

- a. After entry of sperm but before fertilisation
 b. After fertilisation
 c. Before entry of sperm into ovum
 d. Simultaneously with first cleavage

22. The difference between spermiogenesis and spermiation is: (2018)
- In spermiogenesis spermatids are formed, while in spermiation spermatozoa are formed.
 - In spermiogenesis spermatozoa are formed, while in spermiation spermatids are formed.
 - In spermiogenesis spermatozoa from Sertoli cells are released into the cavity of seminiferous tubules, while in spermiation spermatozoa are formed.
 - In spermiogenesis spermatozoa are formed, while in spermiation spermatozoa are released from Sertoli cells into the cavity of seminiferous tubules.
23. Which of the following cells during gametogenesis is normally diploid? (2015)
- Spermatogonia
 - Secondary polar body
 - Primary polar body
 - Spermatid
24. What is the correct sequence of sperm formation? (2013)
- Spermatogonia, Spermatocyte, Spermatid, Spermatozoa
 - Spermatid, Spermatocyte, Spermatogonia, Spermatozoa
 - Spermatogonia, Spermatocyte, Spermatozoa, Spermatid
 - Spermatogonia, Spermatozoa, Spermatocyte, Spermatid

Menstrual Cycle

25. Given below are two statements: One is labelled as **Assertion A** and the other is labelled as **Reason R**. (2024 Re)
- Assertion A:** During menstrual cycle, the ovulation takes place approximately on 14th day.
- Reason R:** Rapid secretion of LH in the middle of menstrual cycle induces rupture of Graafian follicle and thereby the release of ovum.
- In the light of the above statements, choose the most appropriate answer from the options given below.
- A is correct but R is not correct.
 - A is not correct but R is correct.
 - Both A and R are correct and R is the correct explanation of A.
 - Both A and R are correct but R is NOT the correct explanation of A.
26. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R: (2024)
- Assertion A:** FSH acts upon ovarian follicles in female and Leydig cells in male.
- Reason R:** Growing ovarian follicles secrete estrogen in female while interstitial cells secrete androgen in male human being.
- In the light of the above statements, choose the correct answer from the options given below:
- A is true but R is false
 - A is false but R is true
 - Both A and R are true and R is the correct explanation of A.
 - Both A and R are true but R is NOT the correct explanation of A.

27. Which of the following statements are correct regarding female reproductive cycle? (2023)
- In non-primate mammals cyclical changes during reproduction are called oestrus cycle.
 - First menstrual cycle begins at puberty and is called menopause.
 - Lack of menstruation may be indicative of pregnancy.
 - Cyclic menstruation extends between menarche and menopause.
- Choose the most appropriate answer from the options given below:
- A, C and D only
 - A and D only
 - A and B only
 - A, B and C only
28. Which of the following hormone levels will cause release of ovum (ovulation) from the graafian follicle? (2020)
- High concentration of Progesterone
 - Low concentration of LH
 - Low concentration of FSH
 - High concentration of Estrogen
29. A temporary endocrine gland in the human body is: (2017-Delhi)
- Pineal gland
 - Corpus cardiacum
 - Corpus luteum
 - Corpus allatum
30. Changes in GnRH pulse frequency in females is controlled by circulating levels of: (2016 - I)
- Estrogen and progesterone
 - Estrogen and inhibin
 - Progesterone only
 - Progesterone and inhibin
31. Which of the following events is not associated with ovulation in human female? (2015 Re)
- Full development of Graafian follicle
 - Release of secondary oocyte
 - LH surge
 - Decrease in estradiol
32. Menstrual flow occurs due to lack of: (2013)
- Vasopressin
 - Progesterone
 - FSH
 - Oxytocin

Fertilisation and Implantation & Pregnancy and Embryonic Development

33. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R. (2023)
- Assertion A:** Endometrium is necessary for implantation of blastocyst.
- Reason R:** In the absence of fertilization, the corpus luteum degenerates that causes disintegration of endometrium.
- In the light of the above statements, choose the correct answer from the options given below:
- A is false but R is true.
 - Both A and R are true and R is the correct explanation of A.
 - Both A and R are true but R is NOT the correct explanation of A.
 - A is true but R is false.

34. Receptors for sperm binding in mammals are present on: (2021)
 a. Vitelline membrane b. Perivitelline space
 c. Zona pellucida d. Corona radiata
35. Which of the following secretes the hormone, relaxin during the later phase of pregnancy? (2021)
 a. Corpus luteum b. Foetus
 c. Uterus d. Graafian follicle
36. In human beings, at the end of 12 weeks (first trimester) of pregnancy, the following is observed: (2020-Covid)
 a. Most of the major organ systems are formed
 b. The head is covered with fine hair
 c. Movement of the foetus
 d. Eyelids and eyelashes are formed
37. Hormones secreted by the placenta to maintain pregnancy are: (2018)
 a. hCG, hPL, progesterone, prolactin
 b. hCG, hPL, estrogens, relaxin, oxytocin
 c. hCG, hPL, progesterone, estrogens
 d. hCG, progesterone, estrogens, glucocorticoids
38. The amnion of mammalian embryo is derived from (2018)
 a. Ectoderm and mesoderm
 b. Endoderm and mesoderm
 c. Mesoderm and trophoblast
 d. Ectoderm and endoderm
39. Several hormones like hCG, hPL, estrogen, progesterone are produced by: (2016 - II)
 a. Fallopian tube b. Pituitary
 c. Ovary d. Placenta
40. Identify the correct statement on inhibin: (2016 - II)
 a. Inhibits the secretion of LH, FSH and Prolactin
 b. Is produced by granulosa cells in ovary and inhibits the secretion of FSH
 c. Is produced by granulosa cells in ovary and inhibits the secretion of LH
 d. Is produced by nurse cells in testes and inhibits the secretion of LH
41. Fertilisation in humans is practically feasible only if: (2016 - I)
 a. The sperms are transported into vagina just after the release of ovum in fallopian tube
 b. The ovum and sperms are transported simultaneously to ampullary - isthmic junction of the fallopian tube
 c. The ovum and sperms are transported simultaneously to ampullary - isthmic junction of the cervix
 d. The sperms are transported into cervix within 48 hrs of release of ovum in uterus
42. In human females, meiosis-II is not complete until (2015 Re)
 a. Fertilisation b. Uterine implantation
 c. Birth d. Puberty

43. Which of the following layers in an antral follicle is acellular? (2015 Re)
 a. Theca interna b. Stroma
 c. Zona pellucida d. Granulosa
44. Select the correct option describing gonadotropin activity in a normal pregnant female: (2014)
 a. High level of hCG stimulates the thickening of endometrium
 b. High level of FSH and LH stimulates the thickening of endometrium
 c. High level of FSH and LH facilitate implantation of the embryo
 d. High level of hCG stimulates the synthesis of estrogen and progesterone

Parturition and Lactation

45. Match List-I with List-II: (2024 Re)

List-I		List-II	
(A)	Parturition	(I)	Several antibodies for new-born babies
(B)	Placenta	(II)	Collection of ovum after ovulation
(C)	Colostrum	(III)	Foetal ejection reflex
(D)	Fimbriae	(IV)	Secretion of the hormone hCG

Choose the correct answer from the options given below:

- a. A-(III), B-(IV), C-(I), D-(II)
 b. A-(I), B-(IV), C-(II), D-(III)
 c. A-(II), B-(III), C-(IV), D-(I)
 d. A-(III), B-(IV), C-(II), D-(I)
46. Given below are two statements : one is labelled as Assertion A and the other is labelled as Reason R : (2024)
Assertion A: Breast-feeding during initial period of infant growth is recommended by doctors for bringing a healthy baby.
Reason R: Colostrum contains several antibodies absolutely essential to develop resistance for the new born baby.
 In the light of the above statements, choose the most appropriate answer from the options given below :
 a. A is correct but R is not correct.
 b. A is not correct but R is correct.
 c. Both A and R are correct and R is the correct explanation of A
 d. Both A and R are correct but R is NOT the correct explanation of A.
47. Which of these is not an important component of initiation of parturition in humans? (2021, 2015)
 a. Synthesis of prostaglandins
 b. Release of Oxytocin
 c. Release of Prolactin
 d. Increase in estrogen and progesterone ratio

Answer Key

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (c) | 3. (a) | 4. (c) | 5. (a) | 6. (a) | 7. (c) | 8. (b) | 9. (d) | 10. (d) |
| 11. (c) | 12. (c) | 13. (c) | 14. (b) | 15. (d) | 16. (a) | 17. (c) | 18. (c) | 19. (c) | 20. (a) |
| 21. (a) | 22. (d) | 23. (a) | 24. (a) | 25. (c) | 26. (b) | 27. (a) | 28. (d) | 29. (c) | 30. (a) |
| 31. (d) | 32. (b) | 33. (c) | 34. (c) | 35. (a) | 36. (a) | 37. (c) | 38. (a) | 39. (d) | 40. (b) |
| 41. (b) | 42. (a) | 43. (c) | 44. (d) | 45. (a) | 46. (c) | 47. (c) | | | |

